

Trident Compressor Technologies Ltd

ENGINEERING TECHNICAL DATASHEET & COMPLIANCE MATRIX • TENDER REF: CPCL/ENG/2026/HPGC-0412

API 618 5TH EDITION RECIPROCATING GAS COMPRESSOR DATASHEET

Design Specification: Manali Refinery High-Pressure Gas Compression System

Clause 5.4

Engineering Parameter	Tender Specified Mandatory Value	Bidder Offered Value	Compliance Status
Compressor Type & Configuration	API 618 Balanced-Opposed Reciprocating	API 618 5th Ed 2-Crank Balanced Opposed	COMPLIANT
Operating Discharge Pressure	Minimum 120.0 bar(g) continuous	120.0 bar(g) rated discharge	COMPLIANT
Suction Operating Pressure	18.5 bar(g) nominal refinery gas feed	18.5 bar(g) suction rating	COMPLIANT
Driver Motor Rating & Classification	Minimum 1,800 kW, Flameproof Ex d IIB+H2	1,850 kW, 6.6 kV, Ex d IIB+H2 T4	COMPLIANT
Dry Gas Seal Barrier Arrangement	Dual pressurized nitrogen buffer seal	Dual nitrogen buffer barrier seal	COMPLIANT
Vibration Monitoring System	Eddy current proximity probes per API 670	API 670 Vibration & Rod Drop Probes	COMPLIANT

SAFETY INTEGRITY (SIL-3) & AUXILIARY SYSTEM ENGINEERING SCHEDULE

Emergency Shutdown System (ESD), Dry Gas Seals & API 614 Lubrication

Clause 5.4 Schedule B

Subsystem / Module	Engineering Design Standard	Offered Specification	Safety Compliance
Emergency Shutdown System (ESD)	IEC 61508 / 61511 SIL-3 Certification	SIL-3 Certified PLC logic package	CONCORDANT
Lube Oil System	API 614 Package Standards	Forced-feed dual pump lubrication system	CONCORDANT
Pulsation Suppression Dampeners	API 618 Design Approach 3	Acoustic pulsation bottles per API 618 DA-3	CONCORDANT
Skid Structural Baseplate	Rigid structural baseplate	Heavy duty structural steel skid frame	CONCORDANT

CHIEF TECHNICAL OFFICER VERIFICATION & ENGINEERING STAMP

Name: P. Ramaswamy
Designation: Chief Engineer (Turbomachinery & Process Systems)
Entity: Trident Compressor Technologies Ltd
Date & Time: 04-Sep-2026 • Verified Engineering Drawings Attached
Ref / UDIN / Reg: TECH-API618-CONF-0412

